Selective State Space Models for Linear-Time Sequence Modeling (Gu & Dao 2023)

Mamba introduces input-dependent selection mechanisms into structured state space models, achieving linear O(N) sequence scaling, fast inference, and performance matching Transformers on million-length sequences.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

While standard Transformers incur quadratic O(N^2) computational complexity in sequence length N due to full pairwise attention, Mamba introduces input-dependent selection parameters (B, C, Δ) into structured state space models (SSMs). This allows the model to selectively propagate or discard information along sequence trajectories in linear O(N) time with constant O(1) memory state per step during autoregressive generation.

EVIDENCE AND CAUSAL ASSESSMENT

Traditional linear-time recurrent models failed on language because their time-invariant transition matrices could not filter out irrelevant tokens; Mamba's hardware-aware selective scan dynamically adjusts state transitions based on incoming tokens, restoring content-based associative recall.

EFFECT SIZE AND PERSISTENCE

Mamba achieves 5× higher inference throughput than Transformers of comparable parameter size and scales linearly with context length, eliminating the quadratic KV cache memory footprint that bounds Transformer deployments.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as: robust (<20%).

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WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Mamba does not retain infinite sequence memory in lossless form; information is compressed into a fixed-dimensional continuous state vector, meaning historical details undergo lossy decay unless selectively prioritized.

WHAT ACTION IS REASONABLY JUSTIFIED

Evaluate selective state space models (Mamba / SSM hybrids) for production domains requiring long-context stream processing and high-throughput real-time inference without massive KV cache infrastructure.

VERIFIED SOURCES & DOI IDENTIFIERS

- Gu, A., & Dao, T. (2023). Mamba: Linear-Time Sequence Modeling with Selective State Spaces. arXiv preprint arXiv:2312.00752. https://arxiv.org/abs/2312.00752